

OSSP (Operating Systems And Systems Programming) - 25CS2104E

TASK1:

1A). Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that

1. Accept a Linux command as input.
2. Create a child process using fork().
3. Execute the command in the child process using an appropriate exec() system call.
4. Allow the parent process to wait for the child using wait ().
5. Display the Process ID (PID) of both parent and child processes.

1B). Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Part 1: C Program using fork(), exec(), and wait()

Step 1: Create a C file and write the code.

```
sahasra@Mac ~ % nano forkexec.c
```

```
UW PICO 5.09      File: forkexec.c      Modified
{
    printf("\nParent Process\n");
    printf("Parent PID : %d\n", getpid());

    wait(NULL);

    printf("Child process completed.\n");
}

return 0;
}
```

Step 2: Save the file

By pressing Ctrl+O —> Enter —> Ctrl+X

Step 3: Compile the program and Run the program

```
sahasra@Mac ~ % gcc forkexec.c -o forkexec
sahasra@Mac ~ % ./forkexec
Enter Linux command: █
```

```
Enter Linux command: ls

Parent Process
Parent PID : 3017

Child Process
Child PID : 3024
__pycache__          forkexec.c
<!DOCTYPE html>.html from dataclasses import dataclass.py
app.py               import streamlit as st.py
Applications         Internal.html
canteen-preorder     Java.java
CFAIproject          Library
CFAIProject.py        Movies
checkLeapYear(year) {.js Music
const getGrade = student => {.js package-lock.json
default.profrac      Pictures
Desktop              Project.html
DineTechJavaProject.java ProjectSEM2.html
Documents             Public
Downloads             PyCharmMiscProject
fork_exec             st.py
fork_exec.c           streamlit.py
forkexec
Child process completed.
```

Part 2: Operating System Commands

1. Display Operating System Information

```
sahasra@Mac ~ % uname -a
Darwin Mac25.5.0 Darwin Kernel Version 25.5.0: Tue Jun  9 22:26:22 PDT 2026; root:xnu-12377.121.10~1/RELEASE_ARM64_T8132 arm64
```

2. Display CPU Information

```
sahasra@Mac ~ % system_profiler SPHardwareDataType
2026-08-02 19:44:09.972 system_profiler[3044:487464] hw.cpubfamily: 0x6f5129ac
Hardware:
```

Hardware Overview:

```
Model Name: MacBook Air
Model Identifier: Mac16,12
Model Number: MC6T4HN/A
Chip: Apple M4
Total Number of Cores: 10 (4 Performance and 6 Efficiency)
Memory: 16 GB
System Firmware Version: 18000.121.3
OS Loader Version: 18000.121.3
Serial Number (system): LG9NJG793T
Hardware UUID: B2E2436B-6643-5C4C-ABCE-66448F86E00B
Provisioning UDID: 00008132-001928923A7A801C
Activation Lock Status: Disabled
```

3. Display Storage Information

```
sahasra@Mac ~ % diskutil list
/dev/disk0 (internal, physical):
#          TYPE NAME              SIZE       IDENTIFIER
0:         GUID_partition_scheme   *251.0 GB   disk0
1:         Apple_APFS_ISC Container disk1    524.3 MB    disk0s1
2:         Apple_APFS Container disk3    245.1 GB    disk0s2
3:         Apple_APFS_Recovery Container disk2    5.4 GB      disk0s3

/dev/disk3 (synthesized):
#          TYPE NAME              SIZE       IDENTIFIER
0:         APFS Container Scheme -    +245.1 GB   disk3
           Physical Store disk0s2
1:         APFS Volume Macintosh HD   12.6 GB     disk3s1
2:         APFS Snapshot com.apple.os.update-... 12.6 GB     disk3s1s1
3:         APFS Volume Preboot        9.0 GB      disk3s2
4:         APFS Volume Recovery       1.3 GB      disk3s3
5:         APFS Volume Data           108.1 GB    disk3s5
6:         APFS Volume VM             20.5 KB     disk3s6
```

4. Display Running Processes

```
sahasra@Mac ~ % ps
  PID TTY          TIME CMD
 3235 ttys001    0:00.02 -zsh
```

5. Display Live System Usage

```
sahasra@Mac ~ % top
```

```
Processes: 730 total, 3 running, 1 stuck, 726 sleeping, 3036 threads
Load Avg: 3.44, 2.94, 2.79  CPU usage: 11.73% user, 6.43% sys, 81.83% idle
SharedLibs: 1135M resident, 217M data, 147M linkedit.
MemRegions: 0 total, 0B resident, 0B private, 2507M shared.
PhysMem: 15G used (3057M wired, 1055M compressor), 309M unused.
VM: 298T vsize, 6144M framework vsize, 0(0) swapins, 0(0) swapouts.
Networks: packets: 9505208/7909M in, 9250239/7724M out.
Disks: 1266987/23G read, 730945/11G written.
19:46:08
```

PID	COMMAND	%CPU	TIME	#TH	#WQ	#PORT	MEM	PURG	CMPRS	PGRP	PPID	STATE
798	avconference	48.1	12:14.53	48	8	1541+	336M	0B	1856K	798	1	stuck
394	WindowServer	45.2	02:47:06	25/1	6	3262	418M	138M-	58M	394	1	running
527	cameracaptur	24.5	02:50:51	18	4	349-	414M	0B	1184K	527	1	sleeping
0	kernel_task	16.3	01:58:20	618/10	0	0	572M	0B	0B	0	0	running
2971	VTEncoderXPC	11.2	02:11:03	5	2	119+	93M-	0B	912K	2971	1	sleeping
3062	top	5.4	00:01.02	1/1	0	29	8384K+	0B	0B	3062	2995	running
3063	screencaptur	5.1	00:00.44	4	1	217	8976K	752K	0B	611	611	sleeping
3000	com.apple.We	4.4	00:52.53	5	3	67	400M	0B	8080K	3000	1	sleeping
2972	VTDecoderXPC	3.5	01:11.70	6	1	98	59M	0B	816K	2972	1	sleeping
386	bluetoothd	2.8	00:54.53	12	3	430	11M	112K	2000K	386	1	sleeping
412	coreaudiod	2.4	24:07.04	13	4	1269	27M	0B	7792K	412	1	sleeping
530	com.apple.Dr	2.2	22:26.85	7	5	1347	30M	0B	3136K	530	1	sleeping
1102	Safari	1.1	01:37.98	7	2	637	111M	133M	53M	1102	1	sleeping
389	corebrightne	0.9	11:15.89	3	2	190	5200K	0B	1248K	389	1	sleeping
804	AudioConvert	0.8	00:02.45	6	1	65	3968K	0B	608K	804	1	sleeping
749	Spotify Help	0.7	12:28.99	18	2	216	120M	4768K	34M	734	734	sleeping
369	locationd	0.6	00:37.89	6	3	303	10M	240K	2128K	369	1	sleeping